

Hypothesis Testing with E-Values

Chapter 6: The Log-Optimal E-Value and Reverse Information Projection

Based on Ramdas & Wang

Where we are

- Chapter 5: **universal inference** builds a valid e-variable $E^{\text{UI}} = q(Z)/p_{\max}(Z)$ for a composite null $\mathcal{P}$ against a point alternative $\mathcal{Q}$ – easy to construct, but only *asymptotically* log-optimal.
- Chapter 6's question: does a genuinely **finite-sample log-optimal** e-variable exist for composite $\mathcal{P}$? Answer: **yes**, it is unique, it is called the **numeraire**, and it always dominates universal inference.
- The price: the numeraire is characterized through a **duality theory** – the numeraire on one side is paired with a special sub-probability distribution, the **reverse information projection (RIPr)**, on the other.
- Setting fixed for this chapter: composite null $\mathcal{P}$, **point** alternative $\mathcal{Q}$ (mixtures/plug-ins for composite alternatives were Chapter 3.7's business).

Notation you'll need I

Numeraire (plain meaning)

Among all valid e-variables for testing $\mathcal{P}$, the **numeraire** E^* is the single “best” or **reference** e-variable: the one that maximizes worst-case expected log-growth (e-power) under the true data-generating law $\mathbb{Q}$. Think of it as the “gold-standard currency” against which every other e-variable’s performance is measured – that is literally why it is called a numeraire (a reference unit of value, as in economics).

Reverse information projection (RIPr) – plain meaning

The RIPr $\mathbb{P}^*$ is the distribution inside (the closure/bipolar of) the null $\mathcal{P}$ that is “closest” to $\mathbb{Q}$ as measured by Kullback–Leibler (KL) divergence. It is called *reverse* because ordinarily one projects a complicated object onto a simpler model in the argument order $\text{KL}(\text{model}, \text{truth})$; here we project $\mathbb{Q}$ onto $\mathcal{P}$ in the order $\text{KL}(\mathbb{Q}, \mathbb{P})$, with $\mathbb{Q}$ appearing *first* – “reversed” relative to the classical information projection.

Notation you'll need II

Convex duality / strong duality – with an analogy

Convex duality pairs a minimization problem with a maximization problem such that the max is always $\leq$ the min (“weak duality”). **Strong duality** says that under nice conditions (convexity, compactness, etc.) the two values *coincide exactly*.

1-D analogy: imagine two hikers approaching the same mountain pass from opposite sides of a valley. One hiker walks *downhill* from a high peak (minimizing altitude), the other walks *uphill* from the valley floor (maximizing altitude). Weak duality just says the uphill hiker’s altitude never exceeds the downhill hiker’s. Strong duality is the pleasant surprise that, at the pass, both hikers end up at *exactly the same point* – min from one side equals max from the other. In Chapter 6:

$$\underbrace{\max_{E \in \mathfrak{E}} \mathbb{E}^{\mathbb{Q}}[\log E]}_{\text{best e-power, from the “e-variable side”}} = \underbrace{\min_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P})}_{\text{closest null distribution, from the “distribution side”}} .$$

Notation you'll need III

Bayes factor – recap

For simple hypotheses p vs. q , the Bayes factor is the likelihood ratio $B = q(Z)/p(Z)$. For *composite* hypotheses with priors μ (over $\mathcal{Q}$) and ν (over $\mathcal{P}$),

$$B_n = \frac{\int_{\mathcal{Q}} \prod_{i=1}^n q(X_i) \mu(dq)}{\int_{\mathcal{P}} \prod_{i=1}^n p(X_i) \nu(dp)}.$$

B_n is an e-variable *only if* nature itself drew the null from the prior ν – an assumption frequentists are unwilling to make in general. Chapter 6 will show exactly when/how the numeraire coincides with (or beats) B_n .

The set of e-variables $\mathfrak{E}$

Throughout, $\mathfrak{E} = \{E \geq 0 : \mathbb{E}^{\mathbb{P}}[E] \leq 1 \text{ for all } \mathbb{P} \in \mathcal{P}\}$: all nonnegative random variables whose expectation is ≤ 1 under *every* distribution consistent with the null.

6.1 The numeraire e-variable: definition

Intuition first. If we could see the truth $\mathbb{Q}$, we would use the likelihood ratio $d\mathbb{Q}/d\mathbb{P}$ as our e-variable when the null is simple $\mathcal{P} = \{\mathbb{P}\}$ – it is unbeatable (Neyman–Pearson). When $\mathcal{P}$ is composite, is there still a single “best e-variable for every purpose”? Definition 6.1 says: yes, if we require it to beat every other e-variable *in ratio, on average under $\mathbb{Q}$* .

Definition 6.1: numeraire e-variable

A *numeraire* E^* is a $\mathbb{Q}$ -almost-surely strictly positive e-variable such that

$$\mathbb{E}^{\mathbb{Q}} \left[\frac{E}{E^*} \right] \leq 1 \quad \text{for every e-variable } E \in \mathfrak{E}.$$

Symbols: $\mathbb{Q}$ is the (point) alternative; E, E^* are e-variables (nonnegative, $\mathbb{E}^{\mathbb{P}}[E] \leq 1 \ \forall \mathbb{P} \in \mathcal{P}$); the ratio condition says no other e-variable can, on $\mathbb{Q}$ -average, exceed E^* by more than a wash.

Compare to Definition 3.1's ordinary optimality notions: this is a much stronger, “uniformly best in ratio” requirement, not just best in expectation of E itself.

Uniqueness of the numeraire

Step-by-step proof that numeraires are $\mathbb{Q}$ -a.s. unique. Suppose E_1^*, E_2^* are both numeraires. Let $Y = E_2^*/E_1^*$.

- 1 Apply the numeraire property of E_1^* with $E = E_2^*$: $\mathbb{E}^{\mathbb{Q}}[Y] = \mathbb{E}^{\mathbb{Q}}[E_2^*/E_1^*] \leq 1$, so $1/\mathbb{E}^{\mathbb{Q}}[Y] \geq 1$.
- 2 Jensen's inequality (convexity of $x \mapsto 1/x$): $1/\mathbb{E}^{\mathbb{Q}}[Y] \leq \mathbb{E}^{\mathbb{Q}}[1/Y]$.
- 3 Apply the numeraire property of E_2^* with $E = E_1^*$: $\mathbb{E}^{\mathbb{Q}}[1/Y] = \mathbb{E}^{\mathbb{Q}}[E_1^*/E_2^*] \leq 1$.

Chaining: $1 \leq 1/\mathbb{E}^{\mathbb{Q}}[Y] \leq \mathbb{E}^{\mathbb{Q}}[1/Y] \leq 1$. All inequalities are equalities, so Jensen's inequality holds with equality $\Rightarrow Y$ is $\mathbb{Q}$ -a.s. *constant*, and since $\mathbb{E}^{\mathbb{Q}}[Y] = 1$ that constant is 1. Hence $E_1^* = E_2^*$, $\mathbb{Q}$ -almost surely. We may therefore speak of **“the”** numeraire.

Simple null special case

If $\mathcal{P} = \{\mathbb{P}\}$ is simple and $\mathbb{Q} \ll \mathbb{P}$, the numeraire is just the likelihood ratio $E^* = d\mathbb{Q}/d\mathbb{P}$ – exactly the Neyman–Pearson optimal e-variable.

Log-optimality and existence

Proposition 6.3: numeraire $\Leftrightarrow$ log-optimal

A $\mathbb{Q}$ -a.s. strictly positive e-variable E^* is a numeraire if and only if it is *log-optimal*:

$$\mathbb{E}^{\mathbb{Q}} \left[\log \frac{E}{E^*} \right] \leq 0 \quad \text{for every e-variable } E \in \mathfrak{E}.$$

Why: numeraire $\Rightarrow$ log-optimal follows from Jensen (log is concave) applied to the defining ratio inequality. The converse uses a clever one-parameter family $E(t) = (1-t)E^* + tE$ (itself an e-variable for $t \in (0, 1)$) by convexity of $\mathfrak{E}$, a first-order Taylor argument as $t \downarrow 0$, and Fatou's lemma.

Theorem 6.4: existence

A numeraire *always exists*, with no assumptions on $\mathcal{P}, \mathbb{Q}$ whatsoever. It is $\mathbb{Q}$ -a.s. finite if and only if $\mathbb{Q} \ll \mathcal{P}$ (meaning: $\mathbb{Q}(A) > 0 \Rightarrow \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(A) > 0$ for every event A) – a genuinely weak requirement, much weaker than demanding a common dominating measure.

Python: numeraire property on a toy finite $\mathcal{P}$

```
import numpy as np
# P = {P1, P2} two Bernoulli laws (null); Q = Bernoulli(0.8) (alternative).
# Verify  $E^Q[E / E^*] \leq 1$  for several candidate e-variables E.
theta = np.array([0.3, 0.5])          # P1, P2 natural parameters (null)
q_theta, z = 0.8, np.array([0, 1])
def bern_pmf(z, th): return th**z * (1 - th)**(1 - z)
# candidate numeraire = LR of Q to the *largest* null point (6.7.2)
p_star = bern_pmf(z, theta.max())
q_pmf = bern_pmf(z, q_theta)
E_star = q_pmf / p_star
def is_e_variable(E):
    return all(np.sum(bern_pmf(z, th) * E) <= 1 + 1e-9 for th in theta)
candidates = {"E* itself": E_star, "constant 1": np.array([1.0, 1.0]),
              "LR to theta=0.3": q_pmf / bern_pmf(z, 0.3)}
for name, E in candidates.items():
    ratio = np.sum(q_pmf * (E / E_star))
    print(f"{name:18s} e-var={is_e_variable(E)!s:5s}   $E^Q[E/E^*]={ratio:.4f}")$ 
```

6.2 Strong duality for finite $\mathcal{P}$: setup

Intuition. When $\mathcal{P} = \{\mathbb{P}_1, \dots, \mathbb{P}_K\}$ is finite, the set of “achievable null distributions” $\text{Conv}(\mathcal{P})$ (all mixtures $\sum_k \lambda_k \mathbb{P}_k$) is a nice compact convex set (a simplex image). This is exactly the setting where strong duality theorems from convex optimization apply cleanly.

Proposition 6.5: weak duality (always true)

For any e-variable $E \in \mathfrak{E}$ and any $\mathbb{P} \in \text{Conv}(\mathcal{P})$,

$$\mathbb{E}^{\mathbb{Q}}[\log E] \leq \text{KL}(\mathbb{Q}, \mathbb{P}), \quad \text{hence} \quad \sup_{E \in \mathfrak{E}} \mathbb{E}^{\mathbb{Q}}[\log E] \leq \inf_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P}).$$

Derivation: from $\log y \leq y - 1$, a change of measure gives $\mathbb{E}^{\mathbb{Q}}[\log(E \, d\mathbb{P}^a/d\mathbb{Q})] \leq \mathbb{E}^{\mathbb{P}}[E] - 1 \leq 0$ for any $\mathbb{P} \in \text{Conv}(\mathcal{P})$; rearranging gives the displayed inequality. This is the “uphill hiker never above the downhill hiker” half of our analogy.

6.2 Strong duality for finite $\mathcal{P}$: the theorem

Proposition 6.6: strong duality (finite $\mathcal{P}$)

Suppose $\mathcal{P} = \{\mathbb{P}_1, \dots, \mathbb{P}_K\}$, $\mathbb{P}_k \ll \mathbb{Q}$ for all k , and $\min_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P}) < \infty$. Then

$$\mathbb{E}^{\mathbb{Q}}[\log E^*] = \max_{E \in \mathfrak{E}} \mathbb{E}^{\mathbb{Q}}[\log E] = \min_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P}) = \text{KL}(\mathbb{Q}, \mathbb{P}^*),$$

where E^* is the numeraire and $\mathbb{P}^*$ (the RIPr) achieves the minimum.

Why the min is attained. Rewrite $\min_{\mathbb{P} \in \text{Conv}(\mathcal{P})} \text{KL}(\mathbb{Q}, \mathbb{P})$ as $\min_{\lambda \in \Delta_K} \text{KL}(\mathbb{Q}, \sum_{k=1}^K \lambda_k \mathbb{P}_k)$, a strictly convex minimization over the *compact* simplex Δ_K – so a unique minimizer λ^* exists, and $\mathbb{P}^* = \sum_k \lambda_k^* \mathbb{P}_k$.

6.2 Proof of strong duality, step by step

Goal: exhibit *one* e-variable whose e-power equals $\text{KL}(\mathbb{Q}, \mathbb{P}^*)$ – by weak duality that e-variable must then be the numeraire, and equality is forced.

- ① Optimality of $\mathbb{P}^*$ over the convex set $\{\sum_k \lambda_k \mathbb{P}_k : \lambda \in \Delta_K\}$ means for any $\mathbb{P} \in \mathcal{P}$ and any mixing weight $\alpha \in [0, 1]$,

$$\text{KL}(\mathbb{Q}, \mathbb{P}^*) = \min_{\alpha \in [0, 1]} \text{KL}(\mathbb{Q}, \alpha \mathbb{P}^* + (1 - \alpha) \mathbb{P}),$$

i.e. $\alpha = 1$ (pure $\mathbb{P}^*$) already minimizes over this line segment.

- ② First-order (KKT) condition at the boundary optimum $\alpha = 1$: moving from $\alpha = 1$ toward $\alpha = 0$ must not decrease the objective, i.e. the derivative at $\alpha = 1$ is ≥ 0 :

$$-\mathbb{E}^{\mathbb{Q}} \left[\frac{d(\mathbb{P}^* - \mathbb{P})}{d\mathbb{P}^*} \right] \leq 0 \iff \mathbb{E}^{\mathbb{Q}} \left[\frac{d\mathbb{P}}{d\mathbb{P}^*} \right] \leq 1 \iff \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}^*} \right] \leq 1.$$

- ③ The last inequality says exactly: $d\mathbb{Q}/d\mathbb{P}^*$ is an e-variable (its $\mathbb{P}$ -expectation is ≤ 1 for every $\mathbb{P} \in \mathcal{P}$)!
- ④ By definition of e-power, the e-power of this e-variable is $\mathbb{E}^{\mathbb{Q}}[\log(d\mathbb{Q}/d\mathbb{P}^*)] = \text{KL}(\mathbb{Q}, \mathbb{P}^*)$.

Conclusion: the numeraire is $E^* = d\mathbb{Q}/d\mathbb{P}^*$, and its e-power equals $\text{KL}(\mathbb{Q}, \mathbb{P}^*)$ – matching the weak-duality upper bound exactly. This is the “two hikers meet at the pass” moment.

6.3 Reverse information projection: definition

Why we need to go beyond Section 6.2: finiteness of $\mathcal{P}$ gave us compactness of $\text{Conv}(\mathcal{P})$, so the minimizing $\mathbb{P}^*$ was guaranteed to exist *inside* $\text{Conv}(\mathcal{P})$. For general (uncountable, non-dominated) $\mathcal{P}$, this compactness is lost:

- the infimum KL divergence may not be attained inside $\text{Conv}(\mathcal{P})$ at all;
- the true minimizer may be a *sub-probability measure* (total mass < 1), living strictly outside $\text{Conv}(\mathcal{P})$;
- without a common dominating measure, likelihood ratios and KL divergences need care.

Definition 6.7: Reverse information projection (RIPr)

Let E^* be the numeraire for $\mathcal{P}$ against $\mathbb{Q}$. The RIPr of $\mathbb{Q}$ onto $\mathcal{P}$, denoted $\mathbb{P}^*$, is defined *via the numeraire itself*:

$$\frac{d\mathbb{P}^*}{d\mathbb{Q}} = \frac{1}{E^*},$$

understood to be 0 on the event $\{E^* = \infty\}$.

$\mathbb{P}^*$ is unique up to $\mathbb{Q}$ -nullsets, satisfies $\mathbb{P}^* \ll \mathbb{Q}$, and $\mathbb{P}^*(\Omega) \leq 1$ (so it is generally only a *sub-probability measure*) because $\mathbb{E}^{\mathbb{Q}}[1/E^*] \leq 1$.

6.3.1 The effective null is the bipolar of $\mathcal{P}$

Intuition. The *polar* of $\mathcal{P}$ is just its set of e-variables ($\mathcal{P}^\circ = \mathfrak{E}$). Taking the polar *again* (the *bipolar*) gives back the largest family of distributions *indistinguishable from $\mathcal{P}$ using e-variables* – call it the **effective null**.

Polar and bipolar

$$\mathcal{P}^\circ = \{E \in \mathcal{X}_+ : \mathbb{E}^\mathbb{P}[E] \leq 1 \ \forall \mathbb{P} \in \mathcal{P}\} = \mathfrak{E}, \quad \mathcal{P}^{\circ\circ} = \{\mathbb{P} \in \mathcal{M}_+ : \mathbb{E}^\mathbb{P}[E] \leq 1 \ \forall E \in \mathcal{P}^\circ\}.$$

Always $\mathcal{P}^{\circ\circ} \cap \mathcal{M}_1 \supseteq \text{Conv}(\mathcal{P})$, with equality when $\mathcal{P}$ is finite, but the inclusion can be *strict* in general.

Example 6.8 (strict inclusion)

$\mathcal{P}$ = all unit-variance Gaussians with mean < 0 . The standard Gaussian $N(0, 1)$ is *not* in $\text{Conv}(\mathcal{P})$ (it's a boundary/limit point), but it *is* in $\mathcal{P}^{\circ\circ} \cap \mathcal{M}_1$, which in fact contains all unit-variance Gaussians with mean ≥ 0 .

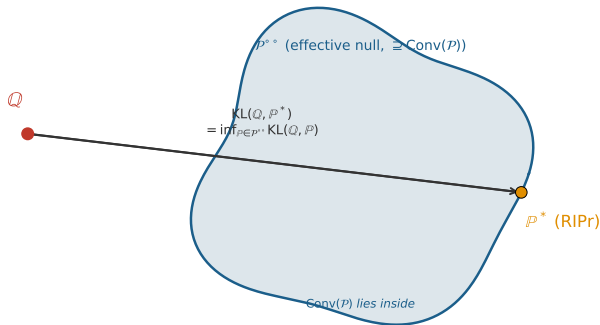
6.3.1 The effective null is the bipolar of $\mathcal{P}$ II

Theorem 6.10

A powered e-variable for $\mathcal{P}$ against $\mathbb{Q}$ exists if and only if $\mathbb{Q} \notin \mathcal{P}^{\circ\circ}$: the bipolar really is the largest set of distributions e-variables cannot distinguish from the null.

6.3 The RIPr picture and general strong duality I

Reverse information projection: Q projected onto the effective null $\mathcal{P}^{\circ\circ}$



$$\text{Numeraire } E^* = \frac{dQ}{d\mathcal{P}^*} \quad \text{and} \quad \text{e-power} = E^Q[\log E^*] = KL(Q, \mathcal{P}^*)$$

6.3 The RPr picture and general strong duality II

Proposition 6.11

The RPr always lies in the bipolar: $\mathbb{P}^* \in \mathcal{P}^{\circ\circ}$.

Theorem 6.13: general strong duality (assuming $\mathbb{Q} \ll \mathcal{P}$)

$$\mathbb{E}^{\mathbb{Q}}[\log E^*] = \sup_{E \in \mathfrak{E}} \mathbb{E}^{\mathbb{Q}}[\log E] = \inf_{\mathbb{P} \in \mathcal{P}^{\circ\circ}} \text{KL}(\mathbb{Q}, \mathbb{P}) = \text{KL}(\mathbb{Q}, \mathbb{P}^*),$$

with no finiteness, dominance, or countability assumptions on $\mathcal{P}$ – the only change from Prop. 6.6 is that the infimum is now over the *bipolar* $\mathcal{P}^{\circ\circ}$ instead of $\text{Conv}(\mathcal{P})$, because that is where the true minimizer may live.

Python: RIPr for a finite $\mathcal{P}$ via convex optimization

```
import numpy as np
from scipy.optimize import minimize
# Finite  $\mathcal{P} = \{P_1, P_2, P_3\}$  on a small sample space,  $Q$  fixed. Solve
#  $\min_{\{\lambda \text{ in simplex}\}} KL(Q, \sum_k \lambda_k P_k) \rightarrow \text{RIPr } P^*.$ 
P = np.array([[0.6, 0.4], [0.3, 0.7], [0.5, 0.5]]) # rows  $P_1, P_2, P_3$ 
Q = np.array([0.2, 0.8])
def kl_objective(raw):
    w = np.exp(raw) / np.sum(np.exp(raw)) # softmax  $\rightarrow$  simplex
    return np.sum(Q * np.log(Q / (w @ P)))
res = minimize(kl_objective, x0=np.zeros(3), method="Nelder-Mead")
lam_star = np.exp(res.x) / np.sum(np.exp(res.x))
P_star = lam_star @ P
E_star = Q / P_star # numeraire =  $dQ/dP^*$ 
print("lambda* :", np.round(lam_star, 4))
print("RIPr P* :", np.round(P_star, 4))
print("E*      :", np.round(E_star, 4))
print("e-power = KL(Q,P*):", round(res.fun, 4))
for k in range(3): # sanity:  $E^{\{P_k\}}[E^*] \leq 1$ 
    print(f"  $E^{P_{k+1}}[E^*] = \{np.sum(P[k]*E_star):.4f\}$ ")
```

6.4 Numeraire is the only likelihood-ratio e-variable

Intuition. The numeraire $E^* = d\mathbb{Q}/d\mathbb{P}^*$ is *by construction* a likelihood ratio of $\mathbb{Q}$ to some element of the effective null. Theorem 6.15's surprise: it is the **only** e-variable with this property. This gives a practical **recipe**: guess a candidate likelihood ratio, verify it's a valid e-variable, and you've automatically found the numeraire.

Theorem 6.15 (verification theorem)

Assume $\mathbb{Q} \ll \mathcal{P}$. Let E^* be $\mathbb{Q}$ -a.s. strictly positive with $\mathbb{P}^*$ defined by $d\mathbb{P}^*/d\mathbb{Q} = 1/E^*$. Then: E^* is the numeraire $\iff \mathbb{P}^* \in \mathcal{P}^{\circ\circ}$. Moreover $\mathbb{E}^{\mathbb{P}^*}[E^*] = 1$ and $\mathbb{P}^*$ is *maximal*: no other $\mathbb{P} \in \mathcal{P}^{\circ\circ}$ equivalent to $\mathbb{Q}$ dominates it setwise.

Corollary 6.16 (practical recipe)

If $\mathcal{P} = \{\mathbb{P} \in \mathcal{M}_1 : \mathbb{E}^{\mathbb{P}}[E] \leq 1 \ \forall E \in \mathfrak{E}_0\}$ for a generating family $\mathfrak{E}_0$, and a candidate $E^* > 0$ satisfies $\mathbb{E}^{\mathbb{Q}}[1/E^*] = 1$ and $\mathbb{E}^{\mathbb{Q}}[E/E^*] \leq 1$ for all $E \in \mathfrak{E}_0$, then E^* is the numeraire.

Lemma 6.17 (shrinking): if $\mathbb{P}^*$ still belongs to the bipolar of a *smaller* null $\overline{\mathcal{P}} \subseteq \mathcal{P}$, the same E^* remains the numeraire for $\overline{\mathcal{P}}$ too – dual to the lifting lemma (6.2) for enlarging $\mathcal{P}$.

6.5 Numeraire dominates universal inference

Intuition. Universal inference builds its e-variable from the *maximum likelihood over the whole null*, evaluated on the actual data: $E^{\text{UI}} = q(Z)/p_{\max}(Z)$ where $p_{\max}(Z) = \sup_{p \in \mathcal{P}} p(Z)$. Since the RPr density p^* is itself just *one* candidate inside (the bipolar of) $\mathcal{P}$, we always have $p^*(Z) \leq p_{\max}(Z)$ – so dividing by the *smaller* quantity p^* gives a *larger* e-variable.

Domination

$$E^* = \frac{q(Z)}{p^*(Z)} \geq \frac{q(Z)}{p_{\max}(Z)} = E^{\text{UI}} \quad (\mathbb{L}\text{-almost surely}).$$

In nondegenerate composite-null cases the inequality is *strict* with positive probability – so universal inference is **inadmissible**: the numeraire is never worse, and is sometimes strictly better, while requiring *no* data splitting and no reference measure at all (unlike UI, which needs densities).

Coherence trade-off: universal inference is *logically coherent* (enlarging the null can only weaken its e-variable); the numeraire generally is *not* coherent. More e-power, less interpretability – a genuine trade-off.

Running example: coin fairness – setting up the numbers

Story. We test whether a coin is fair-or-biased-toward-tails, $H_0 : \theta \leq 0.5$ (a *composite*, “robust” null protecting against any degree of tails-bias), versus the point alternative that the coin has bias $\theta_1 = 0.7$ towards heads. We flip it $n = 20$ times and observe $s = 14$ heads.

Bernoulli exponential family (Section 6.7.1/6.7.2 machinery)

$p_\theta(z) = \theta^z(1 - \theta)^{1-z}$ is a one-dimensional exponential family with sufficient statistic $T(z) = z$ and natural parameter monotone in θ . For a one-sided null $\Theta_0 = (-\infty, \theta_0]$ and point alternative $\theta_1 > \theta_0$, the RPr is p_{θ_0} (the boundary point) and the numeraire is the plain likelihood ratio to that boundary:

$$E^* = \prod_{i=1}^n \frac{p_{\theta_1}(Z_i)}{p_{\theta_0}(Z_i)} = \left(\frac{\theta_1}{\theta_0}\right)^S \left(\frac{1 - \theta_1}{1 - \theta_0}\right)^{n-S}, \quad S = \sum_i Z_i.$$

With $\theta_0 = 0.5$, $\theta_1 = 0.7$, $n = 20$, $S = 14$:

$$E^* = \left(\frac{0.7}{0.5}\right)^{14} \left(\frac{0.3}{0.5}\right)^6 = 1.4^{14} \times 0.6^6 \approx 5.18.$$

Running example: comparing to universal inference I

Universal inference, done the way it is actually used in practice (Chapter 5): split the $n = 20$ flips into a training half (10 flips) and a test half (10 flips). Use the training half only to estimate the boundary of the null (here: since the training data shows more than 50% heads, the restricted MLE sits exactly at the boundary $\theta_0 = 0.5$); accumulate evidence only on the test half:

$$E^{\text{UI}} = \prod_{i \in \text{test}} \frac{p_{0.7}(Z_i)}{p_{0.5}(Z_i)} = \left(\frac{0.7}{0.5}\right)^{s_2} \left(\frac{0.3}{0.5}\right)^{10-s_2}.$$

With $s_2 = 7$ heads out of the 10 test flips (consistent with the same overall 14/20):

$$E^{\text{UI}} = 1.4^7 \times 0.6^3 \approx 2.28.$$

Running example: comparing to universal inference II

Numbers side by side

	Numeraire E^* (full sample)	Split-UI E^{UI} (test half only)
Value	≈ 5.18	≈ 2.28
$\log(\text{value})$, nats	1.646	0.823
Data used	all 20 tosses	10 tosses (10 “spent” on training)

The numeraire is $E^*/E^{\text{UI}} \approx 2.28\times$ larger here – it never throws away data to estimate a training-set boundary, because the RIPr theory handles the composite null for free.

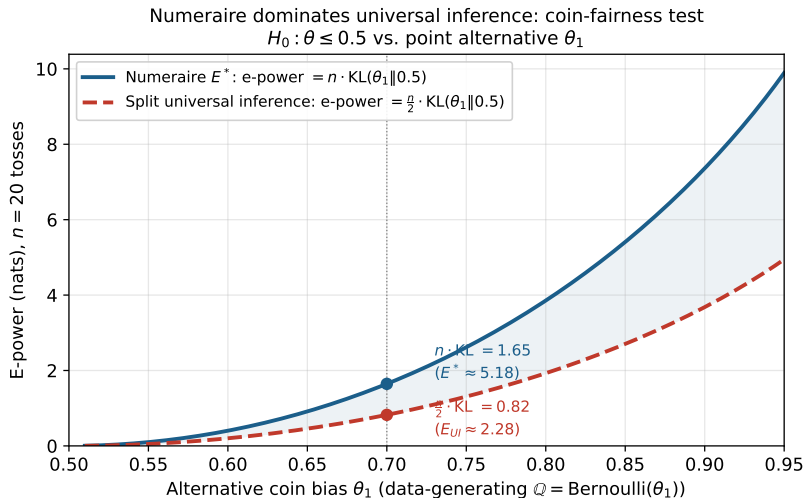
Running example: e-power as a general curve I

In population terms, $\mathbb{E}^{\mathbb{Q}}[\log E^*] = n \cdot \text{KL}(\theta_1 \parallel \theta_0)$ (numeraire uses all n flips) versus $\mathbb{E}^{\mathbb{Q}}[\log E^{\text{UI}}] = \frac{n}{2} \cdot \text{KL}(\theta_1 \parallel \theta_0)$ (split-UI only accrues evidence on the test half), where

$$\text{KL}(\theta_1 \parallel \theta_0) = \theta_1 \log \frac{\theta_1}{\theta_0} + (1 - \theta_1) \log \frac{1 - \theta_1}{1 - \theta_0}.$$

For $\theta_1 = 0.7, \theta_0 = 0.5$: $\text{KL} = 0.0823$ nats/flip, giving the 1.646 vs. 0.823 nats above (a factor of exactly 2, the cost of 50/50 sample-splitting in this illustrative scheme).

Running example: e-power as a general curve II



6.6 E-values and Bayes factors

Intuition. Four quantities compete to compare $\mathcal{Q}$ vs. $\mathcal{P}$ from data $X_1, \dots, X_n$: the Bayesian's Bayes factor, the frequentist's generalized likelihood ratio, universal inference's plug-in, and the numeraire. Only some of these are always valid e-variables.

Four competing statistics

$$B_n = \frac{\int_{\mathcal{Q}} \prod_i q(X_i) \mu(dq)}{\int_{\mathcal{P}} \prod_i p(X_i) \nu(dp)} \quad (\text{Bayes factor, priors } \mu, \nu), \quad G_n = \frac{\sup_{q \in \mathcal{Q}} \prod_i q(X_i)}{\sup_{p \in \mathcal{P}} \prod_i p(X_i)} \quad (\text{generalized LR}),$$

$$U_n = \frac{\int_{\mathcal{Q}} \prod_i q(X_i) \mu(dq)}{\sup_{p \in \mathcal{P}} \prod_i p(X_i)} \quad (\text{universal inference: mixture over } \mathcal{Q}, \text{ max over } \mathcal{P}).$$

- B_n is an e-variable *only if* nature really drew the null from ν – a strong, often unjustified assumption (internal but not external consistency).
- G_n (Wilks' theorem) needs regularity conditions and only gives an *asymptotic* p-value.
- U_n is *always* a valid e-variable, needing only densities – but is generally sub-optimal (Section 6.5).

6.6 The numeraire resolves all of this I

Define $\mathbb{Q}^{\text{eff}} = \int_{\mathcal{Q}} \mathbb{Q}^n \mu(d\mathbb{Q})$ (a mixture alternative) and let $\mathbb{P}^*$ be its RPr onto (the bipolar of) $\{\mathbb{P}^n : \mathbb{P} \in \mathcal{P}\}$. Then

$$E_n = \frac{d\mathbb{Q}^{\text{eff}}}{d\mathbb{P}^*}(X_1, \dots, X_n)$$

is *always* well defined – even with no reference measures at all for $\mathcal{P}, \mathcal{Q}$ – is always a valid e-variable, and it *dominates* U_n .

When do they coincide?

- Both null and alternative simple: B_n, U_n, E_n all coincide.
- Only the null simple: Bayes factor, universal inference, and the numeraire all coincide.
- Neither simple: B_n is only *sometimes* an e-variable; E_n is only *sometimes* representable as a Bayes factor (when the RPr lies inside $\text{Conv}(\mathcal{P})$, i.e. is an actual mixture); G_n is generally neither an e-variable nor a Bayes factor.

The key insight: *for every choice of alternative mixture, there is a unique mixture over the null that reproduces the numeraire as a literal Bayes factor* (Section 6.4) – so the numeraire can be viewed as

6.6 The numeraire resolves all of this II

“the” canonical, uniquely defensible Bayes factor, without needing to justify a subjective prior over the null.

Python: 6.7 examples – computing all the numeraires

```
import numpy as np
# 6.7.1/6.7.2: exp. family, one-sided null  $\theta_1 \leq \theta_0$ , point alt  $\theta_1$ 
def numeraire_exp_family(theta0, theta1, S, n):
    return (theta1/theta0)**S * ((1-theta1)/(1-theta0))**(n-S)
print("Coin fairness E*:", round(numeraire_exp_family(0.5, 0.7, 14, 20), 4))

# 6.7.3: symmetry null;  $E^* = 2 q(z) / (q(z)+q(-z))$ 
def numeraire_symmetry(qz, qmz):
    return 2*qz/(qz+qmz) if (qz+qmz) > 0 else 0.0

# 6.7.4: bounded-mean null  $\{E[Z] \leq \mu\}$ ;  $E^* = 1 + \lambda(Z - \mu)$ , solve FOC
def solve_lambda_star(mu, grid=np.linspace(1e-6, 1/mu-1e-6, 20000)):
    foc = 1 - (1/grid)*np.log((1+grid*(1-mu))/(1-grid*mu))
    return grid[np.argmin(np.abs(foc))]

# 6.7.5: sub-Gaussian null;  $E^* = \exp(\mu Z - \mu^2/2)$ ,  $\lambda = \mu$ 
def numeraire_subgaussian(z, mu):
    return np.exp(mu*z - mu**2/2)
print("Sub-Gaussian E* at z=1.5, mu=1:", round(numeraire_subgaussian(1.5, 1.0), 4))

# 6.7.6: LR-bounded null  $\{dP/dP_0 \leq \gamma\}$ ;  $E^* = (Z \vee z_0)/\gamma$ 
def numeraire_lr_bound(z_star, z0, gamma):
    return max(z_star, z0) / gamma
```

6.7.1–6.7.2 Exponential families & monotone likelihood ratio I

Intuition. For exponential family nulls, restricting the family to its “worst-case boundary point” turns out to give exactly the RIPr – a beautifully concrete instance of the abstract theory.

Setup

$p_\theta(z) = e^{\theta T(z) - A(\theta)}$ w.r.t. reference measure $\mathbb{L}$, A convex/differentiable, T one-dimensional sufficient statistic, natural parameter $\theta \in \Theta$. Null $\mathcal{P} = \{p_\theta : \theta \in \Theta_0\}$ with Θ_0 closed, largest element θ_0 ; alternative $q = p_{\theta_1}$, $\theta_1 > \theta_0$.

Derivation that p_{θ_0} is the RIPr. Using the moment generating function of T ,

$$\mathbb{E}^{\mathbb{Q}} \left[\frac{p_\theta(Z)}{p_{\theta_0}(Z)} \right] = e^{A(\theta_1 + \theta - \theta_0) - A(\theta_1) - A(\theta) + A(\theta_0)}, \quad \theta \in \Theta_0.$$

6.7.1–6.7.2 Exponential families & monotone likelihood ratio II

Any convex f satisfies $f(a + b + c) - f(a + c) \geq f(b + c) - f(c)$ for $a, b \geq 0$; applying this to $f = A$ shows the exponent is ≤ 0 , so the ratio is ≤ 1 for all $\theta \in \Theta_0$ – this is exactly condition (iv) of Theorem 6.14, confirming p_{θ_0} is the RIPr. Hence

$$E^* = \frac{p_{\theta_1}(Z)}{p_{\theta_0}(Z)} = e^{(\theta_1 - \theta_0)T(Z) - A(\theta_1) + A(\theta_0)}.$$

This is exactly our coin-fairness formula, and generalizes it: any one-sided exponential family null makes the numeraire a simple likelihood ratio to the boundary.

6.7.3 Testing symmetry

Intuition. Null: $Z \stackrel{d}{=} -Z$ (a symmetric, entirely non-parametric, non-dominated family). A natural guess for the RIPr is the symmetrization of $\mathbb{Q}$, but because the symmetrized measure need not equal $\mathbb{Q}$ exactly on the support, we must take its *absolutely continuous part*.

RIPr and numeraire for symmetry testing

With q the Lebesgue density of $\mathbb{Q}$,

$$p^*(z) = \frac{1}{2}(q(z) + q(-z)) \mathbb{1}_{\{q(z) > 0\}}, \quad E^* = \frac{d\mathbb{Q}}{d\mathbb{P}^*} = \frac{2q(Z)}{q(Z) + q(-Z)} \mathbb{1}_{\{q(Z) > 0\}}.$$

Why this is the numeraire (sketch). One shows the full set of e-variables here is $\mathfrak{E} = \{E \leq 1 + \phi(Z) : \phi \text{ odd}\}$ (any E of this form has $\mathbb{E}^{\mathbb{P}}[E] \leq 1 + \mathbb{E}^{\mathbb{P}}[\phi(Z)] = 1$ by symmetry). Writing $E^* = 1 + \phi^*(Z)$ with $\phi^*(z) = \frac{q(z) - q(-z)}{q(z) + q(-z)}$ (odd), a direct calculation shows $\mathbb{E}^{\mathbb{Q}}[E/E^*] \leq 1$ for every $E = 1 + \phi(Z) \in \mathfrak{E}$, confirming the numeraire property.

6.7.4 Testing bounded means I

Intuition. Null: $Z \in [0, 1]$ with mean $\leq \mu$ – a fully nonparametric, non-dominated family (contains all point masses on $[0, \mu]$!). Two obvious e-variables are Z/μ and the constant 1; any convex combination $1 + \lambda(Z - \mu)$, $\lambda \in [0, 1/\mu]$, is also one. We search this family for the log-optimal choice.

First-order condition (6.7)

Maximizing $f(\lambda) = \mathbb{E}^{\mathbb{Q}}[\log(1 + \lambda(Z - \mu))]$ (concave in λ) over $\lambda \in [0, 1/\mu]$ gives interior maximizer λ^* solving

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{Z - \mu}{1 + \lambda^*(Z - \mu)}\right] = 0.$$

For $\mathbb{Q} = \text{Uniform}[0, 1]$ this reduces (via direct integration) to the transcendental equation

$$\frac{1 + \lambda^*(1 - \mu)}{1 - \lambda^*\mu} = e^{\lambda^*},$$

solved numerically. The resulting numeraire is $E^* = 1 + \lambda^*(Z - \mu)$, verified via Corollary 6.16 by checking $\lambda = 0$ and $\lambda = 1/\mu$ satisfy the required inequality.

6.7.4 Testing bounded means II

This shows the RPr / numeraire machinery working even when *no* common exponential-family structure is available – only a moment constraint.

6.7.5 & 6.7.6 Sub-Gaussian means and likelihood-ratio bounds

6.7.5: Sub-Gaussian means

Null: Z is 1-sub-Gaussian with nonpositive mean, $\mathcal{P} = \{\mathbb{P} : \mathbb{E}^{\mathbb{P}}[e^{\lambda Z - \lambda^2/2}] \leq 1 \ \forall \lambda \geq 0\}$ (again non-dominated). Alternative $\mathbb{Q} = N(\mu, 1)$, $\mu > 0$. Maximizing $\mathbb{E}^{\mathbb{Q}}[\lambda Z - \lambda^2/2] = \lambda\mu - \lambda^2/2$ over $\lambda \geq 0$ gives $\lambda^* = \mu$, so

$$E^* = e^{\mu Z - \mu^2/2}, \quad \text{RIPr } \mathbb{P}^* = N(0, 1).$$

6.7.6: Testing likelihood-ratio bounds

Null: $\mathcal{P} = \{\mathbb{P} : d\mathbb{P}/d\mathbb{P}_0 \leq \gamma\}$ for fixed $\mathbb{P}_0$, $\gamma \geq 1$. With $Z^* = d\mathbb{Q}/d\mathbb{P}_0$ and $z_0 \geq 0$ the largest constant with $\int_0^{1/\gamma} (q_t(Z^*) \vee z_0) dt = 1$ (q_t = left quantile function), the numeraire is

$$E^* = \frac{Z^* \vee z_0}{\gamma}, \quad \frac{d\mathbb{P}^*}{d\mathbb{P}_0} = \gamma \wedge \frac{\gamma Z^*}{z_0}.$$

The special case $\gamma = 1$ recovers Theorem 3.20's log-optimal e-variable for a simple null, with z_0 the largest constant that is $\leq Z^*$ a.s. (typically $z_0 = 0$).

6.7.7 Testing exchangeability

Intuition. Null: the data vector $Z = (Z_1, \dots, Z_n)$ is *exchangeable* (invariant in distribution under all permutations) – an infinite, non-parametric null with no common dominating measure in general. This connects to permutation/rank tests.

Nuimaire via permutation averaging

Let π be uniform on the symmetric group $\mathcal{S}_n$, and q the density of the (non-exchangeable) alternative $\mathbb{Q}$. Then

$$E = \frac{q(Z)}{q(Z_\pi)} = \frac{q(Z)}{\frac{1}{n!} \sum_{\sigma \in \mathcal{S}_n} q(Z_\sigma)}$$

is an e-variable for the exchangeability null. Because the distribution with density $q(Z_\pi)$ (the permutation-symmetrized density) is itself exchangeable – hence lies in $\mathcal{P}$ – and $d\mathbb{Q}/d\mathbb{P}^*$ is the unique e-variable representable as a bipolar likelihood ratio (Theorem 6.15), E is the numeraire, and $q(Z_\pi)$ is the RPr.

This is a clean illustration of the “guess-and-verify” recipe from Section 6.4: symmetrizing the alternative’s density over all permutations is the natural operation, and duality confirms it is optimal, not just valid.

Chapter 6 summary

- The **numeraire** E^* is the unique, finite-sample log-optimal e-variable for composite $\mathcal{P}$ against point $\mathbb{Q}$; it always exists.
- It is dual to the **reverse information projection** $\mathbb{P}^*$: $E^* = d\mathbb{Q}/d\mathbb{P}^*$, and $\mathbb{E}^{\mathbb{Q}}[\log E^*] = \text{KL}(\mathbb{Q}, \mathbb{P}^*)$ – **strong duality** between maximizing e-power and minimizing KL divergence to the *effective null* $\mathcal{P}^{\circ\circ}$ (the bipolar of $\mathcal{P}$).
- The numeraire is the *only* e-variable expressible as a likelihood ratio to an equivalent element of $\mathcal{P}^{\circ\circ}$ – enabling a practical guess-and-verify recipe (Corollary 6.16).
- It always **dominates universal inference** ($E^* \geq E^{\text{UI}}$), at the cost of losing logical coherence.
- It unifies/subsumes Bayes factors, universal inference, and (asymptotically) the generalized likelihood ratio, coinciding with each under progressively simpler hypotheses.
- Worked out explicitly for: exponential families, monotone likelihood ratio families, symmetry, bounded means, sub-Gaussian means, likelihood-ratio-bounded nulls, and exchangeability.